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# Automorphisms on Catalan Trees and Bracketings

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Two mirror-image isomorphisms between general bracketings and binary bracketings are formalized which parallel the isomorphisms described by De Bruijn and Morselt between plane trees and trivalent plane trees. These maps lead to automorphisms on these four Catalan families. It is found that the orbit structure of these automorphisms is determined by certain "primitive" elements of the four families, and that the enumerator $p_n$ of these primitive elements satisfies the equations: $p_n = m_{n-2} + m_{n-1}$, where $m_n = \sum_{k=0} \binom{n}{2k} c_k$, $c_n$ the Catalan numbers, and

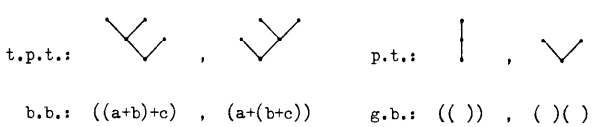
$$c_{n+2} = \sum_{k=0} \binom{n}{k} p_{k+2}.$$

It is then shown that these automorphisms yield isomorphisms between several interesting subsets of the trees and brackets, most of them enumerated by $m_n$. In particular, it is shown that the general bracketings containing no occurrence of "$(g)(g)(g)$" and those containing no occurrence of "$((g))$" are mapped to each other.

## 1. INTRODUCTION

It is well known that the plane trees (p.t.) and the trivalent plane trees (t.p.t.) are equinumerous, and are both enumerated by the Catalan numbers $c_n = (2n)!/n!(n+1)!$ [7, seq. 577]. It is almost as well known that the binary bracketings (b.b.) of $n$ pairs of brackets (including one external pair) about a nonassociative $n+1$-term sum (or product), and the general bracketings (g.b.) built up solely of $n$ pairs of brackets, are also enumerated by the Catalan numbers.

For $n = 2$, for example, these four families are:




(1)

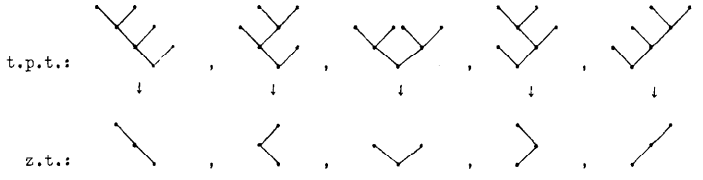


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Two more families, associated with these four, are constructed as follows:

1. Starting with t.p.t. and removing every tip vertex and its covering edge while preserving the (left- or right-) slants of the interior edges produces what I called *zigzag* trees (z.t.) in [2]. These trees are clearly equinumerous with the trivalent trees. For $n = 3$ the correspondence is:




(2)

2. In b.b. the position of the +'s uniquely determines the locations of the terms within the brackets. Hence, deleting the terms while leaving the +'s produces what I will call +.b.b.

Again these brackets are equinumerous with the binary brackets. For $n = 3$ the correspondence is:

$$\begin{array}{lcccc} \text{b.b.:} & (a+(b+(c+d))), & (a+((b+c)+d)), & ((a+b)+(c+d)), & \\ & \downarrow & \downarrow & \downarrow & \\ \text{+.b.b.:} & (+(+(+))), & (+((+)+)), & ((+)+(+)), & \\ & & & ((a+(b+c))+d), & (((a+b)+c)+d) \\ & & & \downarrow & \downarrow \\ & & & ((+(+))+), & (((+)+)+) \end{array} \tag{3}$$

There is a natural correspondence between the two families +.b.b. and z.t.: Each + in +.b.b. corresponds to a vertex of z.t. and the relationship "above" in z.t. corresponds to containedness in the nested parentheses. More precisely, if $(b)$ corresponds to $z$ then

$((b)+)$ corresponds to 

,

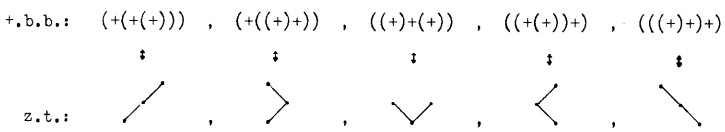
$(+(b))$ corresponds to 

,

and $((b_1)+(b_2))$ corresponds to 

.

For $n = 3$ the correspondence is:




(4)

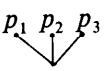
Similarly, there is a natural correspondence between (g.b.)—i.e., g.b. with an extra external pair of brackets—and p.t.: Each pair of facing brackets in (g.b.) corresponds to a vertex of p.t. and the relationship "above" in p.t. corresponds to containedness in the nested parentheses. More precisely, if $(g_i)$ corresponds to $p_i$ then

$((g_1))$ corresponds to 

,

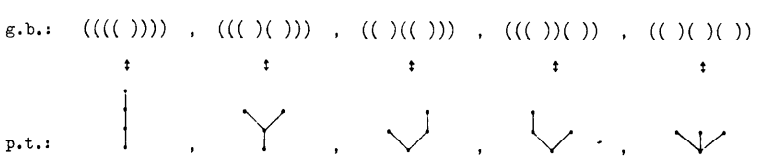
$((g_1)(g_2))$ corresponds to 

,

$((g_1)(g_2)(g_3))$ corresponds to 

,

etc. The extra external pair of brackets in (g.b.) is necessary so that bracketings correspond to trees rather than forests. For $n = 3$ the correspondence is:




(5)

This last correspondence is essentially duplicated in [1] by their "up–down" code, which traces the edges of the trees in clockwise direction.

Replacing the left and right brackets above by $U$'s and $D$'s, respectively, yields the "up–down" codes for the (planted) plane trees.

In Sections II and III two mirror image maps are defined from g.b. to +.b.b. and from p.t. to z.t., and then these are concatenated to define parallel automorphisms on the four families. The orbit structure of these automorphisms is investigated in Section IV.

In Sections V and VI, several subsets of the Catalan families are defined and the effect of restricting the automorphisms to these sets is studied.

## II. The Maps $R$ and $L$

It is known that the distribution of right (closing) brackets in b.b. completely determines the distribution of the left (opening) brackets, and conversely (cf. [4]). This same property holds for +.b.b., as can easily be checked, since each pair of facing brackets contains exactly one + not in an inner bracket. This property of +.b.b. makes it possible to define two mirror image maps from g.b. to +.b.b.:

Definition 1. $R$ (respectively $L$) maps g.b. to +.b.b. as follows:

(1) First replace each right (respectively left) bracket by a +.

(2) Then distribute new right (respectively left) brackets—their positions are uniquely determined.

For $n = 3$, the two correspondences $R$ and $L$ are:

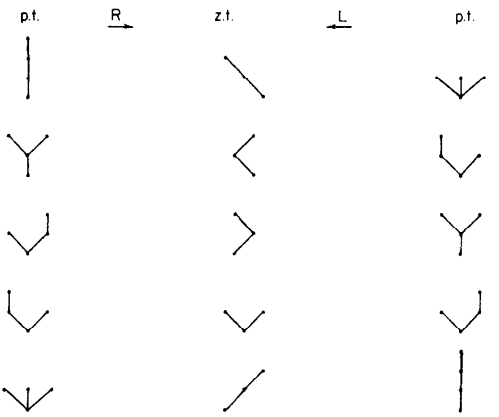
| g.b. | $\overset{R}{\rightarrow}$ | +.b.b. | $\overset{L}{\leftarrow}$ | g.b. |
|---|---|---|---|---|
| ((( ))) | | (((+) +) +) | | ( )( )( ) |
| (( )( )) | | ((+(+)) +) | | (( ))( ) |
| ( )(( )) | | (+((+) +)) | | (( )( )) |
| (( ))( ) | | ((+) +(+)) | | ( )(( )) |
| ( )( )( ) | | (+(+(+))) | | ((( ))) |

(6)

The inverse maps $R^{-1}$ and $L^{-1}$ are defined simply: +.b.b. maps to g.b. under $R^{-1}$ (respectively $L^{-1}$) by removing all right (respectively left) brackets and then changing the +'s into right (respectively left) brackets.

The map from b.b. to g.b. given in [4], which uses the notion of right bracketing, is equivalent to the composite map: b.b. $\rightarrow$+.b.b. $\xrightarrow{R^{-1}}$ g.b.

Since the bracket families correspond naturally to the tree families, (4) and (5), there are two parallel mirror image maps from p.t. to z.t., which I shall also call $R$ and $L$. For $n = 3$, the correspondences are:




(7)

In fact, $R^{-1}$ and $L^{-1}$ from z.t. to p.t. are both defined (in the guise of maps from planted t.p.t. to planted p.t.) in [1].

The basic recursive structure of $R$: g.b. $\rightarrow$ +.b.b. (and, hence, also of its mirror image $L$) is given by

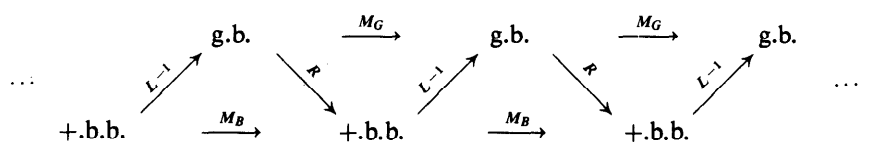
PROPERTY 1. If $R$ maps $g_i$ to $b_i$, $i = 1, 2, 3,...$, then $R$ maps

$$\begin{array}{lll} (g_1), & (g_1)(g_2), & (g_1)(g_2)(g_3), \\ \downarrow & \downarrow & \downarrow \\ (b_1+), & (b_1+(b_2+)), & (b_1+(b_2+(b_3+))), \quad \text{etc.} \end{array}$$

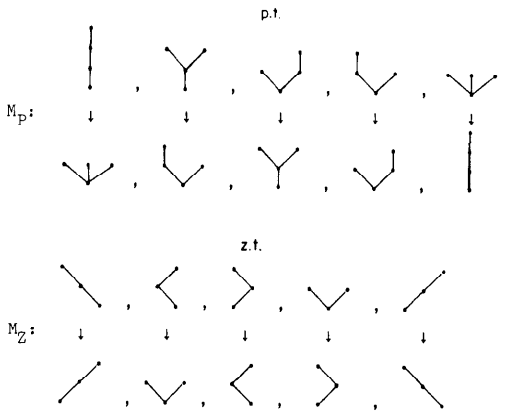
*Proof.* This follows immediately from Definition 1.

## III. THE AUTOMORPHISMS ON g.b., +.b.b., p.t., AND z.t.

The composite maps $M_G = L^{-1} \circ R$ and $M_G^{-1} = R^{-1} \circ L$ are inverse automorphisms mapping g.b. 1–1 onto itself. (See (6), ignoring the middle column.) Similarly, the maps $M_B = R \circ L^{-1}$ and $M_B^{-1} = L \circ R^{-1}$ are inverse automorphisms on +.b.b. These are exhibited in the following commutative diagram:

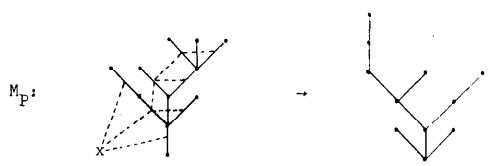



Using the correspondences between the bracketings and trees, (4) and (5), we can define parallel automorphisms $M_P = L^{-1} \circ R$: p.t. $\rightarrow$ p.t. (see (7)) and $M_Z = R \circ L^{-1}$: z.t. $\rightarrow$ z.t. For $n = 3$, for example, the maps $M_P$ and $M_Z$ are:




and (8)

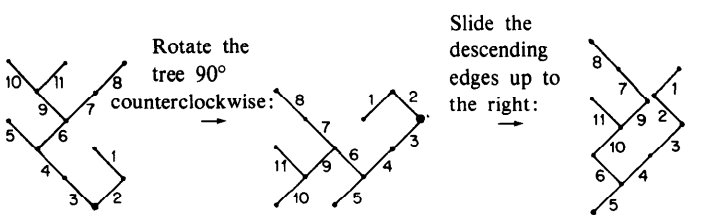
Following the example of [1], the map $M_P$: p.t. → p.t. can be described by the following diagram:




(9)

This map appears in [3, Fig. 13]. Note that the map takes series structure into parallel structure, and conversely.

The map $M_Z$: z.t. → z.t. can also be defined directly. Consider the following diagram, with the edges numbered both for clarity and for future reference:

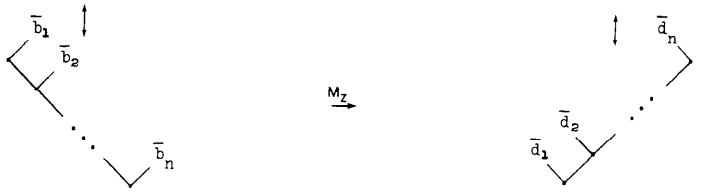



(10)

Note that $M_Z$ takes left slanting edges to right slanting edges, and conversely.

The map $M_Z$ can be defined recursively by use of the following two commutative diagrams, where $g_i \in$ g.b., $b_i$ and $d_i \in$ +.b.b., $L(g_i) = b_i$, $R(g_i) = d_i$, and $b_i$ and $d_i$ correspond to z.t.'s $\overline{b}_i$ and $\overline{d}_i$:

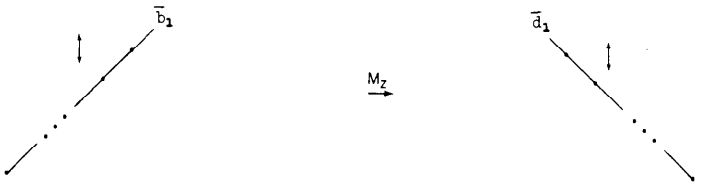
$$(\cdots(+b_1)+b_2)\cdots+b_n) \xrightarrow{L^{-1}} (g_1)(g_2)\cdots(g_n) \xrightarrow{R} (d_1+(d_2+\cdots(d_n+)\cdots)$$




(11)

and

$$(+\cdots(+(+b_1))\cdots) \xrightarrow{L^{-1}} (\cdots((g_1))\cdots) \xrightarrow{R} (\cdots((d_1+)+)\cdots+)$$




(12)

## IV. The Orbit Structure of the Maps

The properties of the four maps $M_G$, $M_B$, $M_P$, and $M_Z$ clearly have much in common. Their orbit structure, in particular, is the same. To talk about the orbit structure of the four Catalan families g.b., +.b.b., p.t., and z.t. under their respective maps, let $M$ represent any of the four maps in what follows.

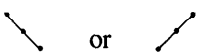
Since $M$ is an automorphism on a finite set, some power of $M$ maps each element to itself. However, $M$ is not a very regular or predictable map, and there appears to be no easy way to tell if any two elements are in the same orbit under $M$, or what the order of any given element is. To give an indication of the chaotic nature of $M$, a polynomial $M_n(x)$ can be formed, for each $n$, in which $a_0 = c_n$ is the number of elements of degree $n$ (i.e., $n$ edges for p.t., $n$ vertices for *z.t.*, and $n$ pairs of brackets for g.b. and +.b.b.), and where, for $i > 0$, $a_i$ counts the number of distinct cycles of order $i$ (so that $\sum_{i=1} i \cdot a_i = a_0$). For $n = 1(1)8$ these polynomials are:

$$\begin{aligned}
M_1(x) &= 1 + x,\\
M_2(x) &= 2 + x^2,\\
M_3(x) &= 5 + x^2 + x^3,\\
M_4(x) &= 14 + x^2 + 2x^6,\\
M_5(x) &= 42 + 2x^2 + 3x^3 + x^5 + 4x^6,\\
M_6(x) &= 132 + 2x^2 + 14x^6 + x^{20} + x^{24},\\
M_7(x) &= 429 + 2x^2 + 7x^3 + 28x^6 + x^9 + x^{10} + x^{15} + 2x^{20}\\
&\quad + x^{24} + x^{48} + x^{72},\\
M_8(x) &= 1430 + 3x^2 + x^4 + 66x^6 + 2x^{10} + 3x^{12} + 2x^{14} + 5x^{20}\\
&\quad + x^{24} + x^{36} + 2x^{42} + x^{48} + x^{54} + 5x^{72} + x^{90} + x^{144}.
\end{aligned} \tag{13}$$

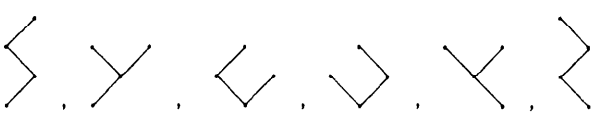
There is, nevertheless, considerable regularity to $M$, and the order and orbit of most elements can be predicted from knowledge of a small set of cycles of what I will call *primitive* elements.

DEFINITION 2. The primitive elements of g.b. are characterized by the absence of either of the two subpatterns "(($g$))", with $g \neq \varnothing$" or ")( )(." The primitive elements of +.b.b. are characterized similarly by the absence of either of the subpatterns "+(+(" or ")+)+."

By the correspondence between +.b.b. and z.t. (4), every instance of one of the two subpatterns "+( +(" and ")+)+" corresponds to a vertex covering exactly two edges of uniform slant; i.e.,



If, following [2], a *branch* of a zigzag tree is defined as the shortest path of uniform slant between branchpoints, where a branchpoint is a tip, root, vertex of degree 3, or vertex of degree 2 connecting edges of opposite slant, then the primitive elements of z.t. are precisely the zigzag trees in which no branch contains more than one edge, so that every vertex is a branchpoint. For $n = 4$, these trees are:



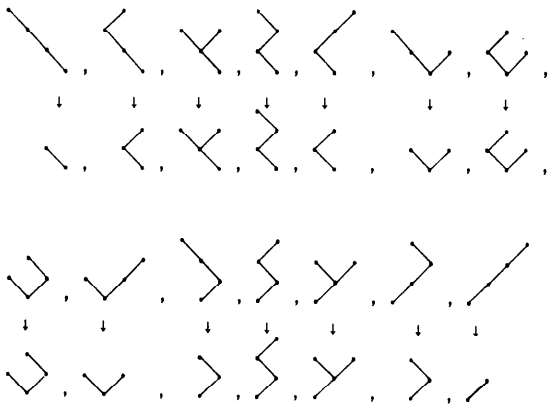
(14)

These trees are called the *branch reduced* zigzag trees in [2], where it is shown that they are enumerated by

$$p_n = m_{n-2} + m_{n-1}, \qquad \text{where} \quad m_n = \sum_{k=0} \binom{n}{2k} c_k. \tag{15}$$

The numbers $m_n = 1, 1, 2, 4, 9, 21, 51, 127, 323, 835,..., n \geqslant 0$, [7, seq. 456], are the Motzkin numbers of [2, 3, 6]. Thus, $p_n = 1, 2, 3, 6, 13, 30, 72, 178, 450, 1158,..., n \geqslant 1$.

In [2] it is further shown that the $c_{n+2}$ z.t.'s with $n+1$ edges are related to the $p_{k+2} = m_n + m_{n+1}$ primitive z.t.'s with $k+1$ edges by the equation $c_{n+2} = \sum_{k=0} \binom{n}{k} p_{k+2}$. In fact, replacing each branch by a single edge collapses $\binom{n}{k}$ trees with $n+1$ edges to each primitive $k+1$-edge z.t. For $n = 4$, this yields:



(16)

Looking at the recursive definition of $M_Z$ given above, (11) and (12), it is clear that $M_Z$ maps branches to branches, in the sense that if the edges are numbered as in (10) above, then the edges of each branch stay together and in order. Hence, if we let $z_p$ be the primitive tree formed from $z$ by collapsing each branch to an edge, then:

PROPERTY 2.

1. $M_Z(z_p)$ is primitive (i.e., $M_Z(z_p) = [M_Z(z_p)]_p$).
2. The diagram

$$\begin{array}{ccc} y & \xrightarrow{M_Z} & z \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ y_p & \xrightarrow{M_Z} & z_p \end{array}$$

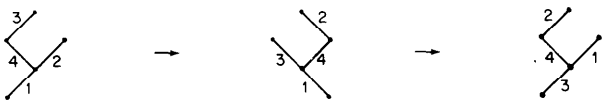
is commutative.

3. $\text{Order}(z_p) \mid \text{Order}(z)$.

Note that 1 says that if an element is primitive, then so is every element in its orbit. Then 2 implies that collapsing every element in the orbit of $z$ yields the elements of the orbit of $z_p$. This together with 3 says that every element in the orbit of $z_p$ is covered the same number of times by the elements in the orbit of $z$.

These properties of $M_Z$ are also properties of the other three maps as well, and they imply that the orbit structure of every element can be determined in part from knowledge of the orbit structure of the primitive elements. In fact, with a little more work it is possible to completely determine the orbit of $z$

once the orbit of $z_p$ is known. To do this it is only necessary to number the edges of $z_p$, as was done in (10) above. For example, $M_Z$ maps:

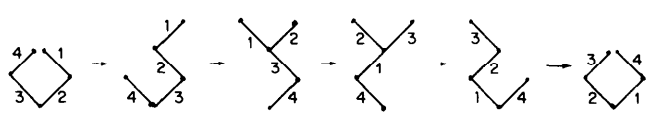


(17)

So while $M_Z^2$ is the identity on the (unnumbered) z.t.'s above, $M_Z^2$ permutes the edges (branches) of the numbered z.t.'s in (17) and the cycle structure of the four edges is (1 2 3)(4). That is, $M_Z^2$ maps edge 1 to the starting position of edge 2, edge 2 to edge 3, and edge 3 to edge 1, while $M_Z^2$ does not move edge 4. The cycle (1 2 3)(4) has order 3 so $(M_Z^2)^3 = M_Z^6$ is the identity of the numbered z.t.'s in (17).

To determine the orbit structure of a given z.t. which collapses to one of the two (unnumbered) primitive trees in (17), we number each branch as we collapse it to an edge according to the number of edges in the branch. If every branch is of different length, we obtain a numbered primitive tree isomorphic to one of the two in (17). However, if various branches have the same number of edges, we obtain a labeled tree with duplicate labels. Depending on which branches are the same length, we may obtain any of the cycle structures (1 2 3)(4), (1 2 3)(1), (1 1 2)(3), (1 1 2)(2), (1 1 2)(1), (1 1 1)(2), or (1 1 1)(1). Since (1 1 1) is of order 1 and (1 1 2) is of order 3, the tree is of order $1 \cdot 2 = 2$ if the cycle structure is either (1 1 1)(2) or (1 1 1)(1), and order $3 \cdot 2 = 6$ otherwise.

As a second example, $M_Z$ maps:



(18)

So while $M_Z^5$ is the identity on the (unnumbered) z.t.'s above, $M_Z^5$ permutes the edges (branches) of the numbered z.t.'s in (18) with cycle structure (1 2 3 4). That is, $M_Z^5$ rotates the four edges of the first tree in (18) above a quarter of a circle clockwise. Every tree collapsing to an unnumbered tree in (18) is isomorphic to a numbered tree with, possibly, duplicate numbers. Hence, for every such tree, $M_Z^5$ has the cycle structure (1 2 3 4), (1 1 2 3), (1 2 1 3), (1 2 1 2), (1 1 2 2), (1 1 1 2), or (1 1 1 1). Since (1 2 3 4), (1 1 2 3), (1 2 1 3), (1 1 2 2), and (1 1 1 2) are of order 4, (1 2 1 2) is of order 2, and (1 1 1 1) is of order 1, the tree must be of order $4 \cdot 5 = 20$, $2 \cdot 5 = 10$, or $1 \cdot 5 = 5$.

The order and cycle structure of the primitive elements appears not to

have the slightest regularity, and no general statements about their behavior appears possible. Table I gives the structure of the primitive elements for $n = 1(1)8$.

TABLE I

| $n$ | $p_n$ | Order of $z_p$ | Cycle structure of $z_p$ | Possible orders of $z$ |
|---|---|---|---|---|
| 1 | 1 | 1 | none | none |
| 2 | 2 | 2 | (1) | 2 |
| 3 | 3 | 3 | (1 2) | 3, 6 |
| 4 | 6 | 6 | (1)(2)(3) | 6 |
| 5 | 13 | 5 | (1 2 3 4) | 5, 10, 20 |
| | | 3 (2 cycles) | (1 2)(3 4) | 3, 6 |
| | | 2 | (1 2 3)(4) | 2, 6 |
| 6 | 30 | 24 | (1 2 3)(4 5) | 24, 48, 72, 144 |
| | | 6 | (1)(2)(3)(4)(5) | 6 |
| 7 | 72 | 24 | (1 2 3)(4 5 6) | 24, 72 |
| | | 18 | (1 2 3 4)(5 6) | 18, 36, 72 |
| | | 15 | (1 2 3 4 5 6) | 15, 30, 45, 90 |
| | | 9 | (1 2 3 4 5 6) | 9, 18, 27, 54 |
| | | 3 (2 cycles) | (1 2)(3 4)(5 6) | 3, 6 |
| 8 | 178 | 42 (2 cycles) | (1 2 3)(4)(5)(6)(7) | 42, 126 |
| | | 14 (2 cycles) | (1 2)(3)(4)(5)(6)(7) | 14, 28 |
| | | 12 (3 cycles) | (1 2 3 4)(5 6 7) | 12, 24, 36, 48, 72, 144 |
| | | 10 (2 cycles) | (1 2 3)(4 5)(6 7) | 10, 20, 30, 60 |
| | | 6 | (1)(2)(3)(4)(5)(6)(7) | 6 |
| | | 4 | (1 2 3 4 5)(6 7) | 4, 8, 20, 40 |

Since z.t. corresponds to +.b.b. (4), and +.b.b. can be mapped to g.b. (6), which in turn corresponds to p.t. (5), the cycle structure of the primitive z.t.'s is the same as that of the other three Catalan families. Thus, the order and orbit of any element of +.b.b., g.b., or p.t. can be determined by mapping it to the corresponding element $z$ of z.t. and then looking at the cycle structure of the underlying primitive element $z_p$ with numbered edges (branches).

An element of +.b.b. is collapsed to a primitive element by replacing each occurrence of "+( +($b$))" by "+($b$)" and each occurrence of "(($b$) +) +" by "($b$)+." For $n = 3$, for example, we have:

$$\begin{array}{ccccccccc} (((+)+)+) & & ((+(+))+) & & (+((+)+)) & & ((+)+(+)) & & (+(+(+))) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ ((+)+) & & ((+(+))+) & & (+((+)+)) & & ((+)+(+)) & & (+(+)) \end{array} \tag{19}$$

An element of g.b. is collapsed to a primitive element by replacing each occurrence of "$((g))$, with $g \neq \varnothing$" by "$(g)$" and each occurrence of "$)($ $)($" by "$)($." For $n = 3$, for example, we have:

$$\begin{array}{ccccccccc} (\ )(\ )(\ ) & & ((\ ))(\ ) & & ((\ )(\ )) & & (\ )((\ )) & & (((\ ))) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ (\ )(\ ) & & ((\ ))(\ ) & & ((\ )(\ )) & & (\ )((\ )) & & ((\ )) \end{array} \tag{20}$$

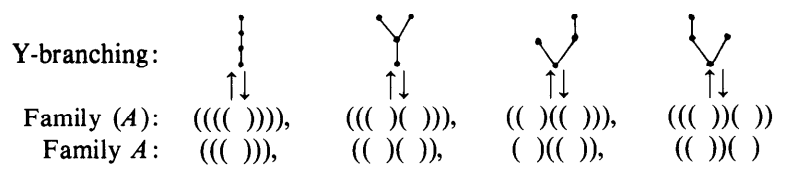
## V. Motzkin Families

Two families of plane trees, which are equinumerous and enumerated by the Motzkin numbers $m_n$ (15), have appeared in [2, 3]. These are the *Y-branching* plane trees, in which every vertex is of degree at most three and the root is of degree at most two (see (21) below), and the *branch-reduced* plane trees, in which no vertex, the root excepted, has degree 2 (see (22) below). Under the correspondence between p.t. and (g.b.), these two families correspond to two equinumerous families of brackets, which I shall call families $(A)$ and $(G)$:

Definition 3. Family $A$ contains all g.b.'s with no occurrence of the subpattern "$(g)(g)(g)$." Family $(A)$, i.e., $A$ with an extra external pair of brackets, corresponds to the Y-branching p.t.'s.

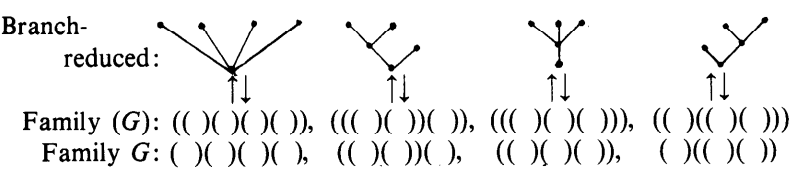
Definition 4. Family $G$ contains all g.b.'s with no occurrence of the subpattern "$((g))$." Family $(G)$, i.e., $G$ with an extra external pair of brackets, corresponds to the branch reduced p.t.'s. In both of these definitions the $g$'s may be vacuous.

For $n = 3$ these families are:




(21)

and




(22)

These families are of interest here because despite the generally chaotic behavior of $M_P$ and $M_G$, when the maps are restricted to these families they become remarkably tame. In fact, $M_P^3$ maps the planted Y-branching p.t.'s 1–1 onto the branch-reduced p.t.'s, and conversely. Similarly, $M_G^3$ maps $(A)$ and $G$ 1–1 onto each other, as I will show below. As a consequence, $M^6$ is an automorphism on these families, but it is not the identity. In fact, there are elements in these families of at least the orders 1, 2, 3, 4, 6, 9, 12, 20, 24, 36, 42, 48, 54, 72, and 126.

To demonstrate that $M_G^3$ maps $(A)$ and $G$ 1–1 onto each other, it will be necessary to break $M_G^3$ down into the sequence of maps $L^{-1} \circ R \circ L^{-1} \circ R \circ L^{-1} \circ R$, where the maps $L$ are $R$ were given in Definition 1 in Section II above. Applying the maps $R$ and $L^{-1}$ in succession, starting with family $(A)$, leads to the following cycle:

$$\begin{array}{ccccccccccc} (A) & \xrightarrow{R} & (B+) & \xrightarrow{L^{-1}} & (C(\ )) & \xrightarrow{R} & D & \xrightarrow{L^{-1}} & E & \xrightarrow{R} & F & \xrightarrow{L^{-1}} & G \\ \| & & & & & & & & & & & & \| \\ (A)^{\perp} & \xleftarrow{L^{-1}} & (+B^{\perp}) & \xleftarrow{R} & (\ )C^{\perp}) & \xleftarrow{L^{-1}} & D^{\perp} & \xleftarrow{R} & E^{\perp} & \xleftarrow{L^{-1}} & F^{\perp} & \xleftarrow{R} & G^{\perp} \end{array} \tag{23}$$

where the families $A$ through $G$ are defined by:

| | |
|---|---|
| $A$(g.b.): no $(g)(g)(g)$ | $B$(+.b.b.): no ))) |
| $C$(g.b.): no ))) | $D$(+.b.b.): no )+) |
| $E$(g.b.): no )( ) | $F$(+.b.b.): no +(+ |
| $G$(g.b.): no $((g))$ | |

and families $B^{\perp}$, $C^{\perp}$, $D^{\perp}$, $E^{\perp}$, and $F^{\perp}$ are the mirror images of $B$ through $F$.

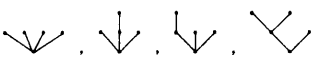
I will show that the map from $(A)$ to $G$ carries the complement of each family into the complement of the next by showing that each pattern of constraints implies the next. This implies that the complement of $(A)$ in g.b. is mapped into the complement of $G$. The proof is then completed by noting that $R$ and $L^{-1}$ are 1–1 and that the complements of $(A)$ and $G$ are equinumerous because $(A)$ and $G$ are equinumerous with the Y-branching and branch-reduced plane trees, which are shown in [2, 3] to be equinumerous.

| Map | Pattern of constraints |
|---|---|
| $R$:$(A) \to (B+)$: | $(g)(g)(g) \to (b+(b+(b+)))$ by Property 1, Section II |
| $L^{-1}$:$(B+) \to (C(\ ))$: | $))) \to ))))$ by the definition of $L^{-1}$ |
| $R$:$(C(\ )) \to D$: | $))) \to +++ \to +) \cdots )+) \cdots )+$ since every two +'s are separated when the new right brackets are added |
| $L^{-1}$: $D \to E$: | $)+) \to )(\ )$ by the definition of $L^{-1}$ |
| $R$: $E \to F$: | $)(\ ) \to +(+$ by the definition of $R$ |
| $L^{-1}$: $F \to G$: | $+(+ \Rightarrow (b+(+b)) \to g((g))$ |

The subsets of p.t. corresponding to the subsets $(C(\ ))$ and $E$ of g.b. defined above are:

$T_{(C(\ ))}$:

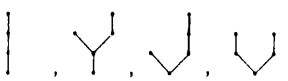
The rightmost edge above each vertex must terminate:



(24)

$T_{(E)}$:

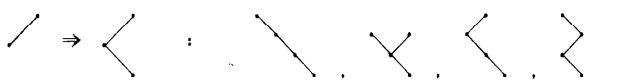
Only the leftmost edge above each vertex can terminate:



(25)

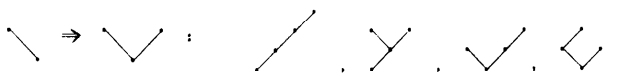
Similarly, the subsets of z.t. corresponding to the subsets $(B+)$, $D$ and $E$ of +.b.b. defined above are:

$T_{(B+)}$:



(26)

$T_D$:



(27)

$T_F$:



(28)

Note the similarly of family $T_D$ to the Y-branching plane trees in (21). Ignoring the slant in family $T_D$, these are the same set of trees.

## VI. Four Intersection Families

There are four interesting subfamilies of families $A$ through $G$ and their mirror images families $A^{\perp}$ through $G^{\perp}$ that can be formed by intersecting pairs of families. If we define family $H$ by $H = (A) \cap G$, then $H$ contains all g.b.'s with no occurrence of the subpatterns "$(g)(g)(g)$" or "$((g))$." Hence every nonempty pair of brackets in $H$ must contain "$((g_1)(g_2))$," the $g_i$'s possibly vacuous. Then the p.t.'s corresponding to $(H)$ are the trivalent plane trees (t.p.t.), which are enumerated by the Catalan numbers. Similarly, in the intersection $D \cap D^{\perp}$ every pair of brackets must contain "$(+)$" or "$((b_1) + (b_2))$," so the subset of z.t. that this family corresponds to is also t.p.t. Note that $M_G^3$ maps $(A) \cap G$ 1–1 onto itself and $M_B^3$ maps $D \cap D^{\perp}$ 1–1 onto itself.

The two families $A$ and $G$ are symmetric in that $A = A^{\perp}$ and $G = G^{\perp}$ while

families $(C(\ ))$ and $E$ are not symmetric. However, the two intersection families $J = (C(\ ))\cap((\ )C^{\perp})$ and $K = E \cap E^{\perp}$ are symmetric. Since in the above cycle of bracket families the map $M_G$ commutes with taking intersections, it follows that $M_G^3$ maps families $J$ and $K$ 1–1 onto each other. Hence, these two families are equinumerous. Family $J$ is characterized by the absence of the subpatterns "(((" and "))),"  and family $K$ is characterized by the absence of the subpatterns ")(  )" and "(  )(." If we let $a_n$ enumerate these two families, then it can be demonstrated that $a_n = 1, 1, 2, 4, 8, 17, 37, 82, 185, 423, 978,...$ satisfies the equation:

$$a_n = \sum_{k=1} \frac{1}{n-k}\binom{n-k}{k}\binom{n-k}{k-1} = \sum_{k=1} d_{n-k,k}, \tag{29}$$

where $d_{n,k} = n^{-1}\binom{n}{k}\binom{n}{k-1}$ is shown in [5, 6] to be the number of ballot paths with $k$ horizontal segments, and hence the number of plane trees with $k$ tips (vertices other than the root of degree one).

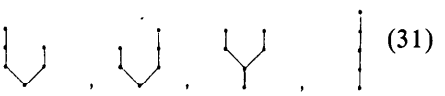
The plane trees corresponding to families $J$ and $K$ can be seen from the definitions of families $T_{(C(\ ))}$ and $T_{(E)}$ to be:

$T_{((\ )C\cap C^{\perp}(\ ))}$: The outermost edges above each vertex must terminate:



(30)

$T_{(E\cap E^{\perp})}$: The vertex below each tip is of degree two:



(31)

The above formula (29) for $a_n$ follows from the observation that the $d_{n,k}$ $n+k$-edge trees in $T_{(E\cap E^{\perp})}$ with $k$ tips can be formed from the $d_{n,k}$ plane trees with $n$ edges and $k$ tips by adding an extra edge at each tip.

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